

AITA-INTELLIGENT — Software Requirements Specification (SRS)

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1. Introduction

1.1. Document Purpose

This Software Requirements Specification (SRS) document describes all functional and non-functional requirements of the **AITA-Intelligent** system (AI-Powered Teaching Assistant & AST Code Analytics Platform). This document serves as the foundation for system design, development, testing, and acceptance within the scope of the SWP391 – Software Development Project course.

1.2. Project Scope

AITA-Intelligent is an intelligent programming education support platform, consisting of:

- **Web Portal (PWA-ready)** for students and lecturers (Mobile Native is Future Scope).
- **Automated code grading system** via a custom-built, self-hosted Docker sandbox (Code Execution Engine).
- **Generative AI module** (GenAI) supporting assignment creation, Clean Code grading, and compilation error explanation.
- **Source code plagiarism detection tool** based on Abstract Syntax Tree (AST) analysis combined with the Winnowing algorithm.
- **Background processing queue** (Redis Queue) and team contribution analysis via Git Analytics.

1.3. Intended Audience

Audience	Purpose
Instructor	Product evaluation and acceptance
Development Team (Dev Team)	Design and development reference
P2P Peer Reviewers	Requirement completeness review
Defense Committee	Final evaluation

1.4. Glossary

Term	Definition
AST	Abstract Syntax Tree – A tree representation of the logical structure of source code
Winnowing	An algorithm for creating digital fingerprints from k-gram sequences for text/code matching
Sandbox	An isolated code execution environment implemented via Docker containers with team-configured resource limits (RAM, CPU, network, PID, filesystem)
JWT	JSON Web Token – A stateless authentication standard
SSO	Single Sign-On – One-click login via Google OAuth 2.0
BullMQ	A Redis-based queue library for Node.js
LLM	Large Language Model – e.g., GPT-4o, Gemini
ERD	Entity-Relationship Diagram
SRS	Software Requirements Specification
P2P	Peer-to-Peer – Cross-team evaluation
UAT	User Acceptance Testing
CI/CD	Continuous Integration / Continuous Deployment
LOC	Lines of Code
k-gram	A contiguous subsequence of k elements, used in the Winnowing algorithm
Fingerprint	Digital fingerprint – A set of hash values representing a document/source code

1.5. References

- Schleimer, S., Wilkerson, D.S., Aiken, A. (2003). *"Winnowing: Local Algorithms for Document Fingerprinting"*. ACM SIGMOD 2003.
- Parr, T. (2010). *"Language Implementation Patterns"*. Pragmatic Bookshelf.
- Sommerville, I. (2016). *"Software Engineering"*, 10th Edition. Pearson.
- Docker Engine API Documentation - <https://docs.docker.com/engine/api/>
- OpenAI API Reference – <https://platform.openai.com/docs/api-reference>

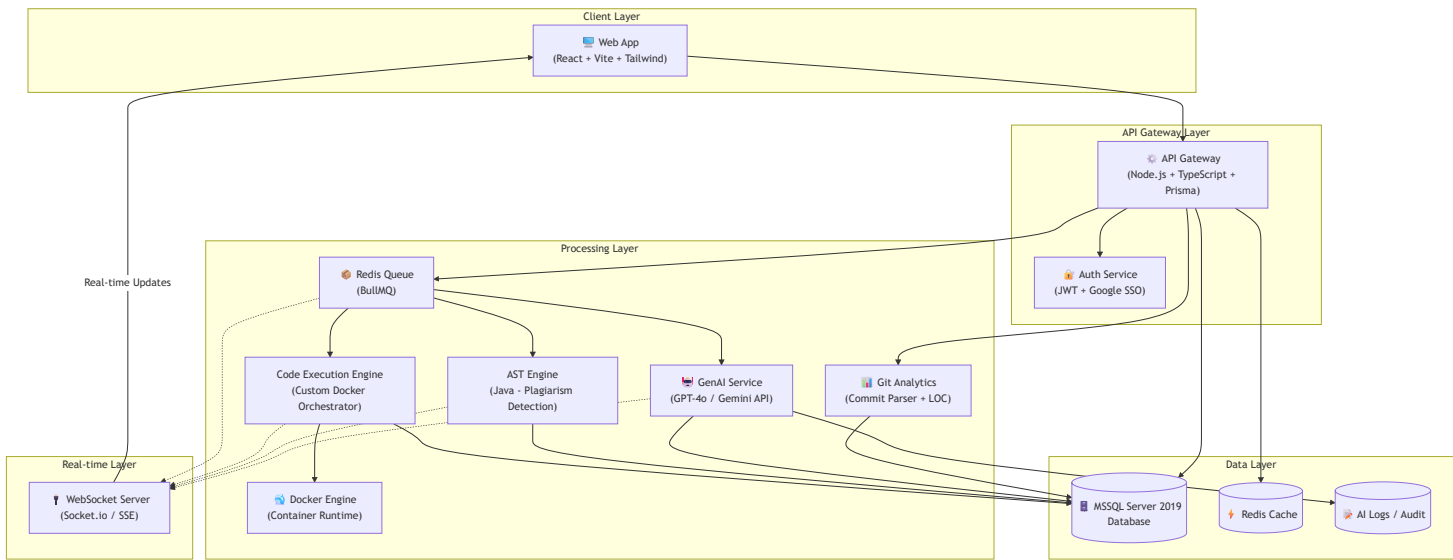
2. System Overview

2.1. Product Context

The AITA-Intelligent system addresses real-world challenges in programming education at universities:

- **Time-consuming manual grading:** Lecturers spend hours grading code for large classes. AITA automates this process using Code Execution Engine.
- **Sophisticated source code fraud:** Students use AI (ChatGPT, Copilot) to write submissions and then disguise them (rename variables, reorder functions). Traditional text comparison tools are ineffective. AITA uses AST + Winnowing for detection.
- **Lack of quality feedback:** Students only see scores without understanding their mistakes. AITA integrates LLM to explain errors in natural language.

2.2. High-Level Architecture



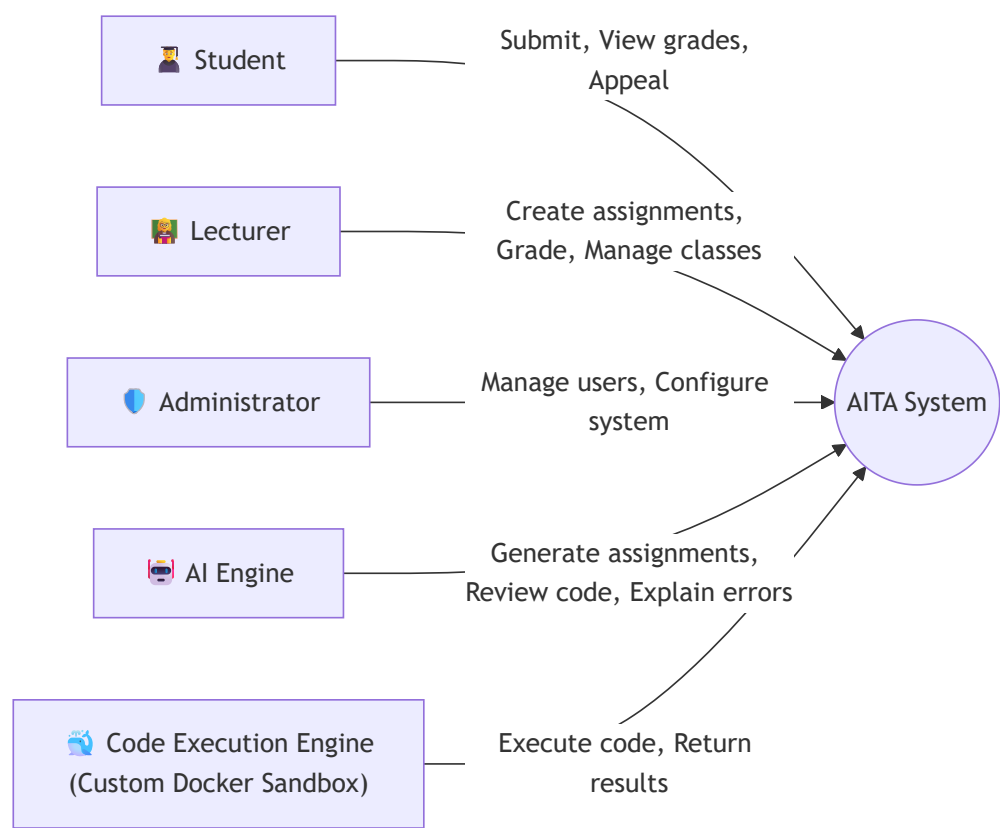
2.3. Core Subsystems

#	Subsystem	Description	Key Technology
1	Portal & Auth	User interface, authentication, authorization	React, Vite, Tailwind, JWT, Google OAuth 2.0
2	Code Execution Engine	Automated code grading in self-managed isolated Docker containers	Docker Engine API (Dockerode), Node.js
3	GenAI Core & Review Hub	Assignment generation, Clean Code grading, error explanation	OpenAI GPT-4o / Gemini API, LangChain
4	AST Plagiarism Detection	Source code plagiarism detection using AST + Winnowing	ANTLR4 (Java grammar), FastAPI wrapper

#	Subsystem	Description	Key Technology
5	Redis Queue & Git Analytics	Background processing queue, team contribution analysis	BullMQ, Redis, Git CLI Parser

3. System Actors

3.1. Actor Diagram



3.2. Detailed Actor Descriptions

Actor	Role	Key Permissions
Student	Students use the system to submit assignments, view grading results, and appeal if they disagree with the outcome	SSO login (@fpt.edu.vn), view assignment list, upload .zip file, view real-time results, submit appeal
Lecturer	Lecturers create classes, design exams, manage grading rubrics, monitor results, and handle complaints	Create/edit/delete classes, import student lists from Excel, create assignments + test cases, request AI-generated assignments/rubrics, view statistical reports, handle appeals, view Git Analytics

Actor	Role	Key Permissions
Administrator	System administrators manage accounts, configure Sandbox settings, and monitor system operations	Manage users, configure Sandbox (RAM, CPU limits), manage API keys (OpenAI/Gemini), view audit logs
AI Engine	Automated system actor – Not a human. Performs content generation and code quality evaluation tasks	Generate assignments from lecturer descriptions, generate rubrics, evaluate Clean Code (SOLID, naming, architecture), explain compilation errors in natural language
Code Execution Engine	Automated system actor. An isolated runtime that executes student-submitted source code	Create an isolated Docker container per submission, compile and execute code inside it, enforce resource limits (RAM/CPU/network/PID) and timeout directly via Docker Engine API, capture stdout/stderr, then destroy the container

4. User Stories

4.1. Student Features

ID	User Story	Acceptance Criteria	Priority
US-01	As a student , I want to log in using my FPT Google account (@fpt.edu.vn / @fe.edu.vn) so I don't need to create a new account and my identity is verified.	- Only accept FPT domain emails - Redirect to Dashboard after login - JWT stored in HttpOnly Cookie	● High
US-02	As a student , I want to view the list of Assignments for my enrolled class, so I can track deadlines and submission status.	- Display assignment name, deadline, status (Not submitted / Submitted / Graded) - Sort by nearest deadline	● High
US-03	As a student , I want to submit assignments as a .zip file containing source code, so the system can automatically grade them for me.	- Only accept .zip files (max 10MB) - File hashed with SHA-256 for integrity verification - System displays "Submission received" immediately - Reject submission with error message if student has already submitted ≥ 5 times for this assignment (see R-01)	● High
US-04	As a student , I want to view grading results in real-time (test case scores, Clean Code score, plagiarism check results) as soon as the system finishes processing, without needing to reload the page.	- Real-time updates via WebSocket - Clearly display: Passed/Failed for each test case - Display Clean Code score with AI explanation	● High

ID	User Story	Acceptance Criteria	Priority
		- Display plagiarism similarity percentage (0-30% Safe, 30-60% Warning, >60% Danger/Flag)	
US-05	As a student , I want to submit an Appeal if I believe the automated grading result is inaccurate, so the lecturer can re-evaluate.	- Appeal form with "Reason" field (textarea) - Appeal status: Pending / Accepted / Rejected - Student receives notification when lecturer responds	 Medium

4.2. Lecturer Features

ID	User Story	Acceptance Criteria	Priority
US-06	As a lecturer , I want to create a new class and import student lists from an Excel file , to save time on manual data entry.	- Upload .xlsx file, system auto-parses columns: Student ID, Full Name, Email - Use DB Transaction to ensure full import or rollback on error - Report successful/failed import counts	 High
US-07	As a lecturer , I want to create Assignments with test cases (StdIn/StdOut format), so the system can automatically grade student submissions.	- Input title, description, deadline - Add multiple test cases (each with: Input, Expected Output, score weight) - Select programming language for Sandbox (Java, Python, C#)	 High
US-08	(Simplified – MVP) As a lecturer , I want to request AI to generate a basic assignment draft from my brief description, to reduce assignment preparation time.	- Lecturer enters a prompt describing the topic (e.g., "Assignment on linked list") - AI returns: basic assignment description and sample test cases - Lecturer must review and finalize before saving - MVP Note: Full rubric auto-generation is a stretch goal	 Medium
US-09	As a lecturer , I want to view a Dashboard with overview statistics (submission rate, score distribution, plagiarism suspects) for each assignment, to quickly assess class performance.	- Pie chart: submission rate - Bar chart: score distribution - Table: Top 10 submission pairs with highest similarity % (AST Plagiarism)	 High

ID	User Story	Acceptance Criteria	Priority
US-10	As a lecturer , I want to handle student appeals (Accept or Reject with reason), to ensure fairness in grading.	- List of Pending appeals - Review code + original grading results - Accept (re-grade) or Reject (with reason) button	 Medium
US-11	<i>(Simplified – MVP)</i> As a lecturer , I want to view basic Git Analytics for each team member (commits, LOC), to get a high-level view of individual contributions.	- Input GitHub Repo URL of the team - Display: Total commits, LOC added/removed per member - <i>MVP Note: PR analysis and contribution timeline charts are planned for future iterations</i>	 Medium

4.3. System / AI / Code Execution Features

ID	User Story	Acceptance Criteria	Priority
US-12	As the system , when receiving a .zip submission file, I must enqueue the submission into Redis (BullMQ) for asynchronous processing, to avoid blocking the main API thread.	- Submission enqueued with metadata: submissionId, studentId, language - Queue has retry mechanism (max 3 retries) on job failure - API immediately returns status 202 Accepted	 High
US-13	As the Code Execution Engine , when receiving a job from Redis Queue, I must execute student code in a self-managed isolated Docker container with limits: 512MB RAM, limited CPU, disabled network access (--network=none), run with test cases, then return results.	- Sandbox auto-cleans after completion or timeout (30 seconds) - Code Execution Engine enforces via Docker: process isolation (namespaces), resource limiting (cgroups: --memory , --cpus , --pids-limit), security (seccomp profile, read-only filesystem, non-root user) - Return results: Passed/Failed for each test case + stdout/stderr	 High
US-14	As the AST Engine system , after code execution finishes, I must convert source code to an AST, remove surface-level disguises (variable names, comments, function order), apply Winnowing to create fingerprints, and compare against all other submissions in the same assignment to calculate similarity %.	- Primary language: Java (ANTLR parser) - <i>Python/C# support planned for future iterations</i> - Remove: variable names, function names, comments, whitespace, function order - Winnowing: k-gram size = 25, window size = 40 - Store fingerprints in ASTFingerprints	 High

ID	User Story	Acceptance Criteria	Priority
		table - Create Similarity Matrix: all submission pairs	
US-15	As the AI system (LLM) , after the Code Execution Engine finishes grading, I must evaluate source code quality (Clean Code, SOLID, naming convention, layer architecture) and explain compilation errors in Vietnamese natural language so students can understand.	- Use GPT-4o or Gemini API (exponential backoff retry) - Chain-of-Thought + Few-Shot Learning prompts - Return: Clean Code score (0-100), specific comments per file, compilation error explanation if any - Timeout: max 60 seconds/request	Medium

5. Detailed Use Case Specifications

UC-01: Student Submits Assignment (Submit Assignment)

Attribute	Description
Primary Actor	Student
Secondary Actors	Code Execution Engine, AST Engine, AI Engine
Preconditions	Student is logged in, Assignment deadline has not passed, Student has submitted fewer than 5 times for this assignment
Main Flow	1. Student selects Assignment from the list 2. Student uploads <code>.zip</code> file containing source code 3. System validates file (checks size $\leq 10\text{MB}$, <code>.zip</code> format) 4. System hashes file with SHA-256 for integrity verification 5. System creates <code>Submission</code> record with <code>PENDING</code> status 6. System pushes job to Redis Queue (<code>BullMQ</code>) 7. Redis Queue dispatches job to Worker 8. Code Execution Engine creates an isolated Docker container, compiles and executes the code inside it, then captures test case results 9. System updates status to <code>GRADING</code> 10. AST Engine parses source code, creates fingerprint, checks for plagiarism 11. AI Engine evaluates Clean Code + explains errors 12. System aggregates final score; updates status to <code>COMPLETED</code> if similarity $\leq 60\%$, or <code>SCORE_WITHHELD</code> if similarity $> 60\%$ (pending lecturer review) 13. Sends real-time results to Student via WebSocket
Alternative Flow	4a. File is corrupt or exceeds size limit $\rightarrow$ Display error, request resubmission 8a. Student code contains malware (fork bomb, <code>/proc</code> access, disk abuse) $\rightarrow$ Code Execution

Attribute	Description
	Engine detects malicious signal via Docker/seccomp → Status <code>SECURITY_VIOLATION</code> 8b. Code fails to compile → Return compilation error + AI error explanation 8c. Code execution exceeds 30 seconds without malicious signal → Status <code>TIMEOUT</code> , notify student to optimize algorithm 10a. Similarity detected > 60% → Flag as <code>PLAGIARISM_DETECTED</code> , notify lecturer
Exception Flow	7a. Redis Queue is full or down → System retries 3 times, if still failing → Status <code>QUEUE_ERROR</code> , notify Admin 11a. OpenAI/Gemini API timeout → Skip Clean Code score, grade based on test cases + AST, mark "AI Review Pending"
Postconditions	<code>Submission</code> record is updated with final status and score. Student receives results on the interface.

UC-02: Lecturer Creates Assignment with AI (Create Assignment with AI)

Attribute	Description
Primary Actor	Lecturer
Secondary Actors	AI Engine
Preconditions	Lecturer is logged in, has created at least 1 class
Main Flow	- Lecturer selects class and clicks "Create New Assignment" - Lecturer enters: Title, General Description, Programming Language, Deadline (Optional) Lecturer clicks "Generate with AI": enters a topic description prompt - AI Engine returns: Detailed assignment, sample Input/Output, Grading Rubric - Lecturer reviews and edits AI-generated content - Lecturer adds Test Cases (StdIn → Expected StdOut) manually or from AI - Lecturer clicks "Save & Publish" - System creates Assignment record, sends notification to all students in the class
Alternative Flow	3a. Lecturer doesn't use AI → Manually enters the entire assignment 4a. AI returns unsuitable results → Lecturer clicks "Regenerate" with a different prompt 6a. Test cases are invalid (Expected Output is empty) → System warns and requests correction
Postconditions	Assignment is created and displayed in the class assignment list.

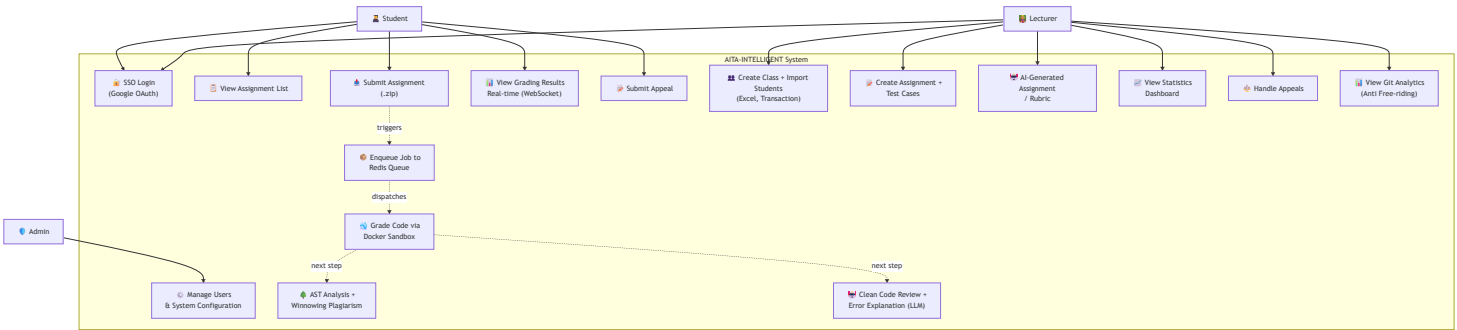
UC-03: Lecturer Imports Student List (Bulk Import Students)

Attribute	Description
Primary Actor	Lecturer
Preconditions	Lecturer has created a class, has a properly formatted Excel file

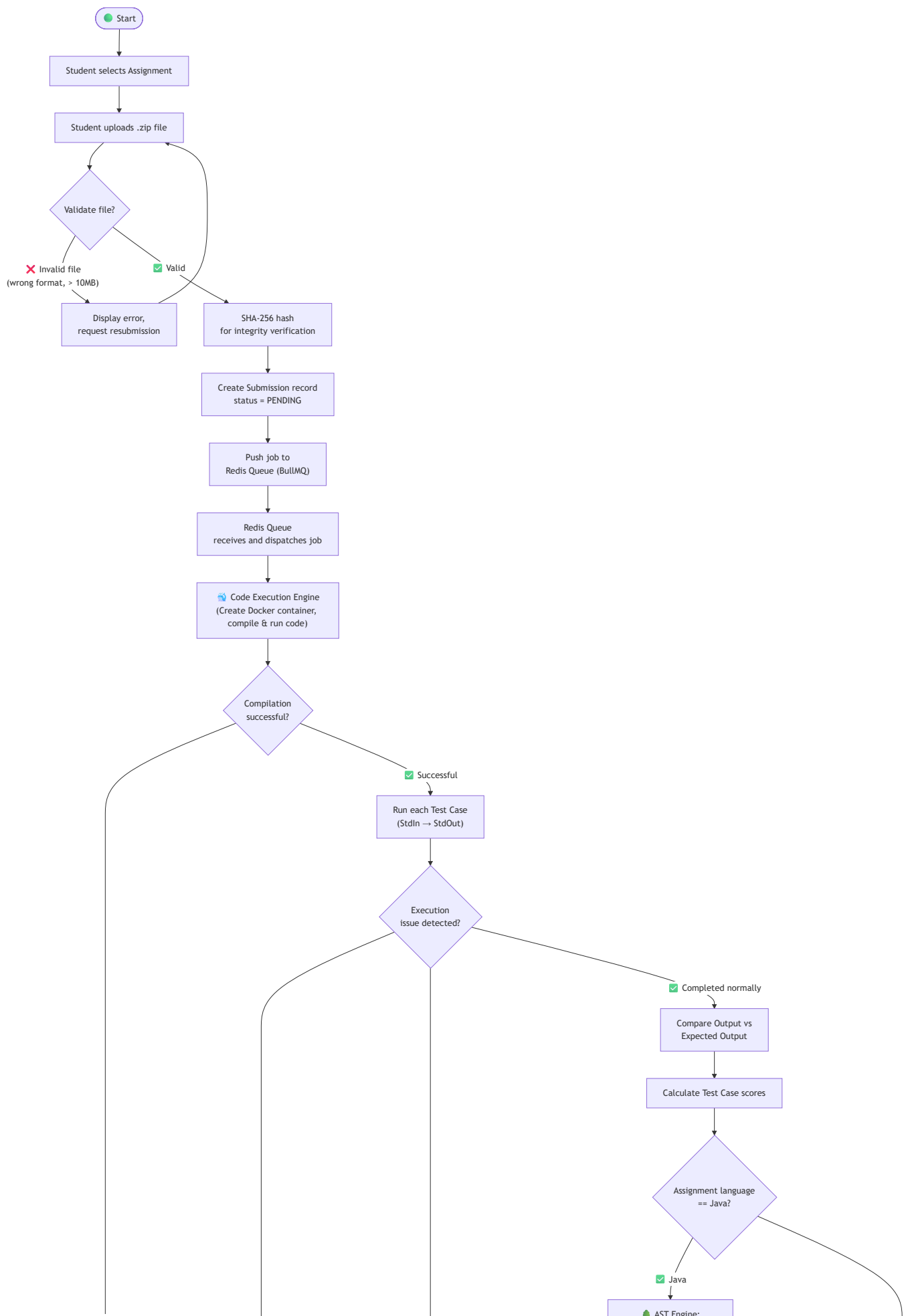
Attribute	Description
Main Flow	1. Lecturer selects class, clicks "Import Students" 2. Lecturer uploads .xlsx file 3. System parses file, extracts: Student ID, Full Name, Email 4. System displays preview list (for lecturer to verify) 5. Lecturer clicks "Confirm Import" 6. System uses SQL Transaction to insert all students 7. Displays results: X students successful, Y duplicate records
Alternative Flow	3a. Excel file has wrong format (missing columns) → Display detailed error, request re-upload 6a. Transaction fails (e.g., duplicate email) → Rollback entirely, report specific duplicate rows
Postconditions	Student list is added to the class. New accounts are created (if not already existing).

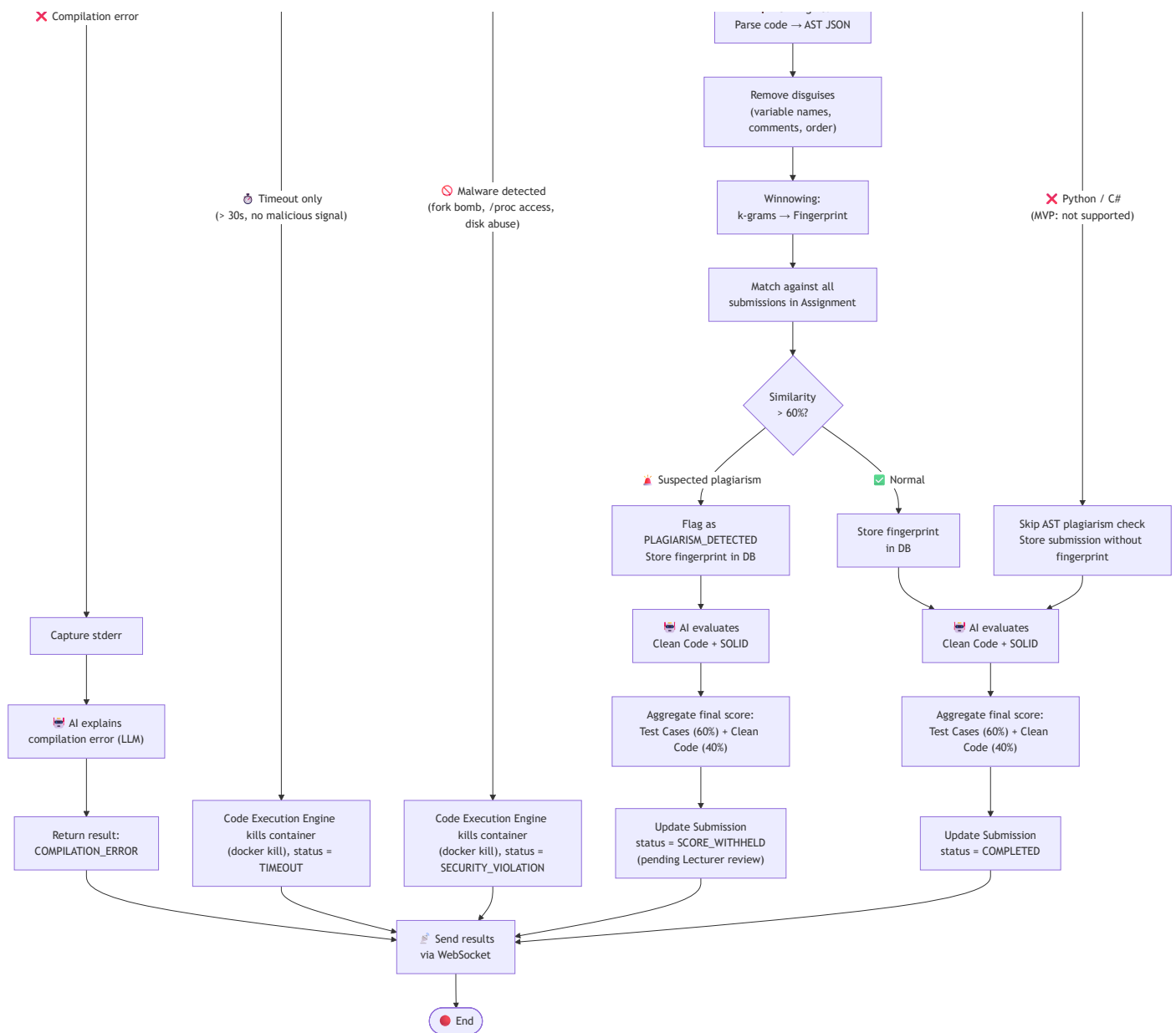
6. UML Diagrams

6.1. Use Case Diagram (Overview)

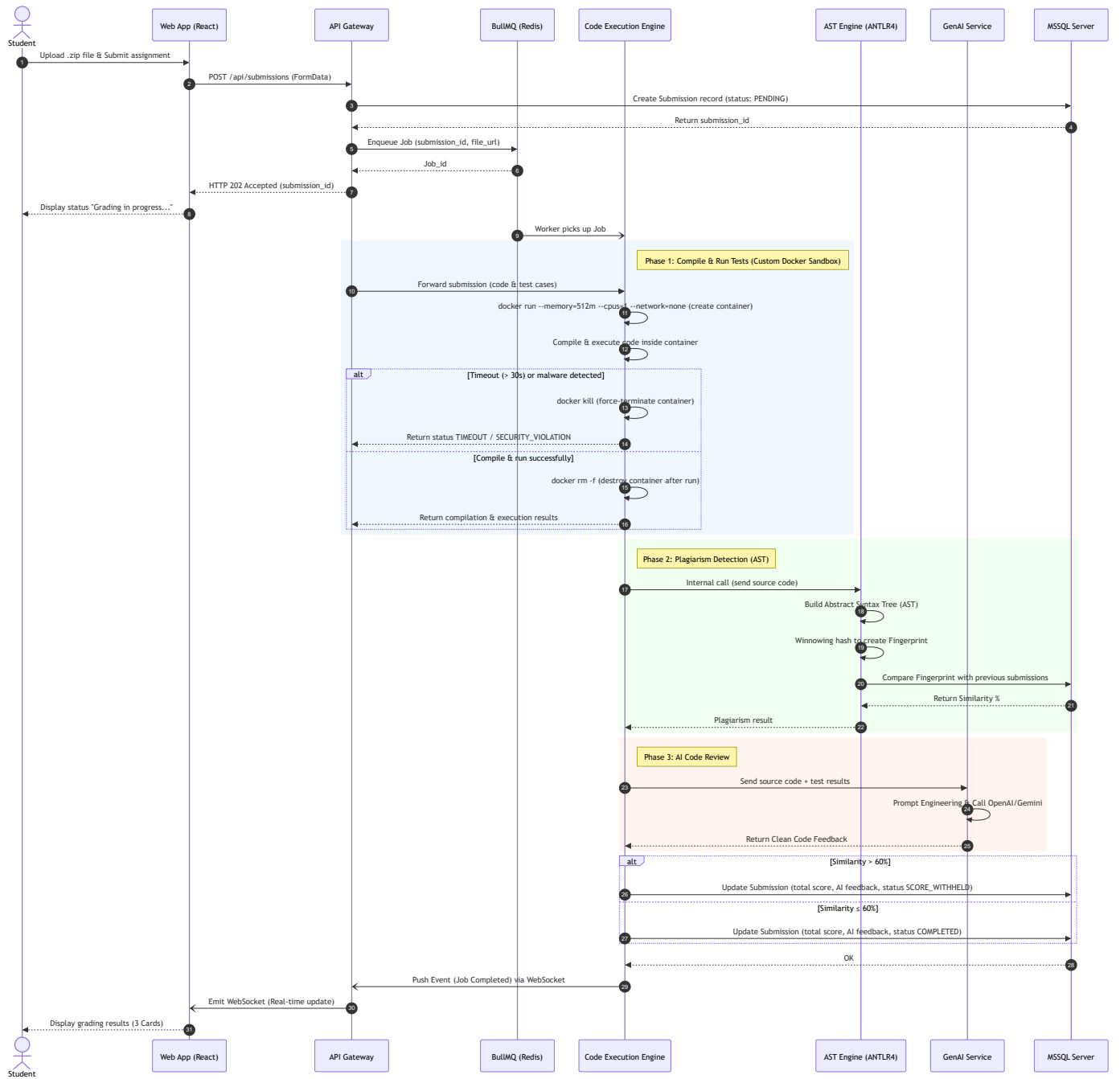


6.2. Activity Diagram – Submission and Automated Grading Flow



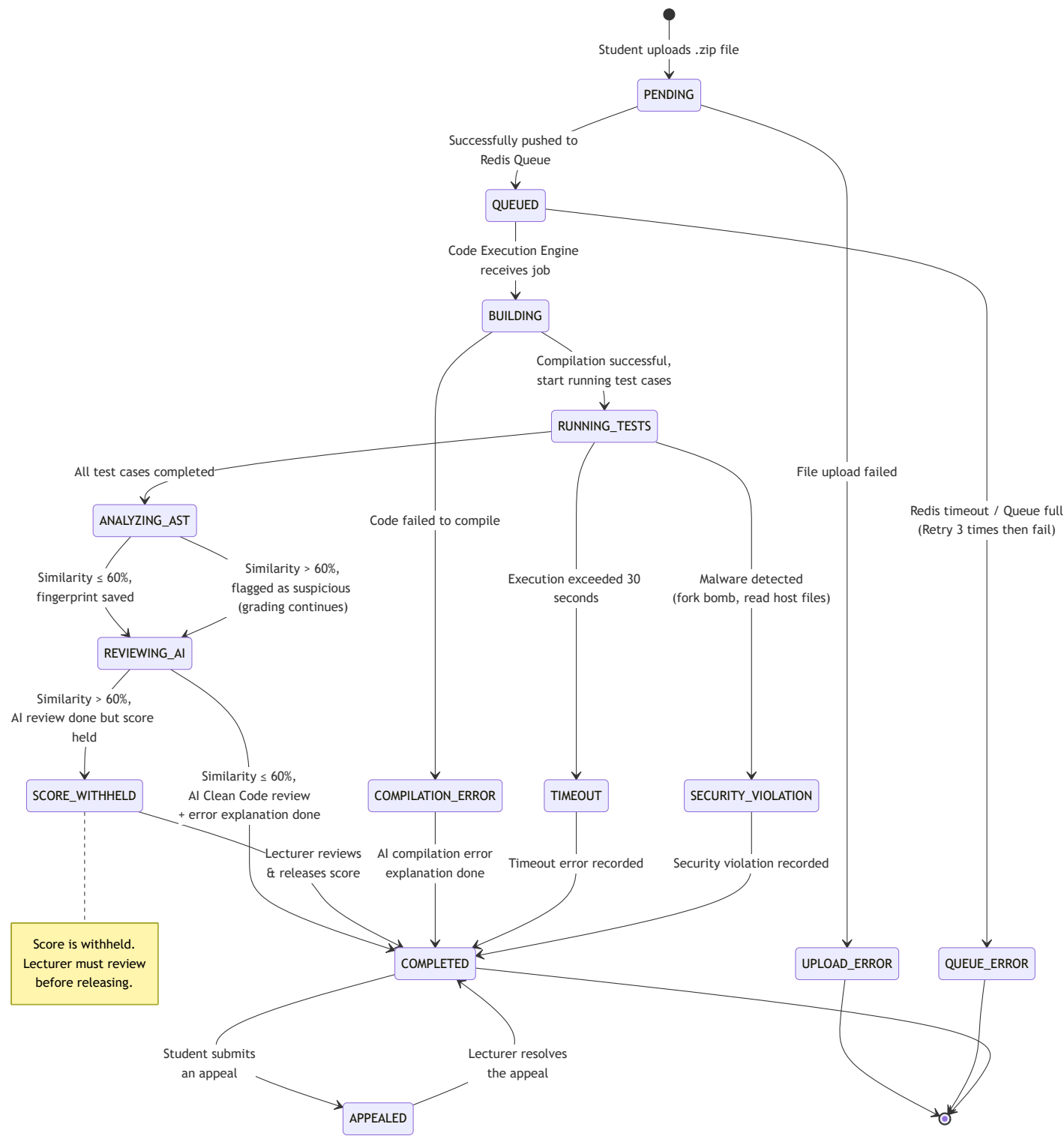


6.3. Sequence Diagram – Submission Grading API Flow

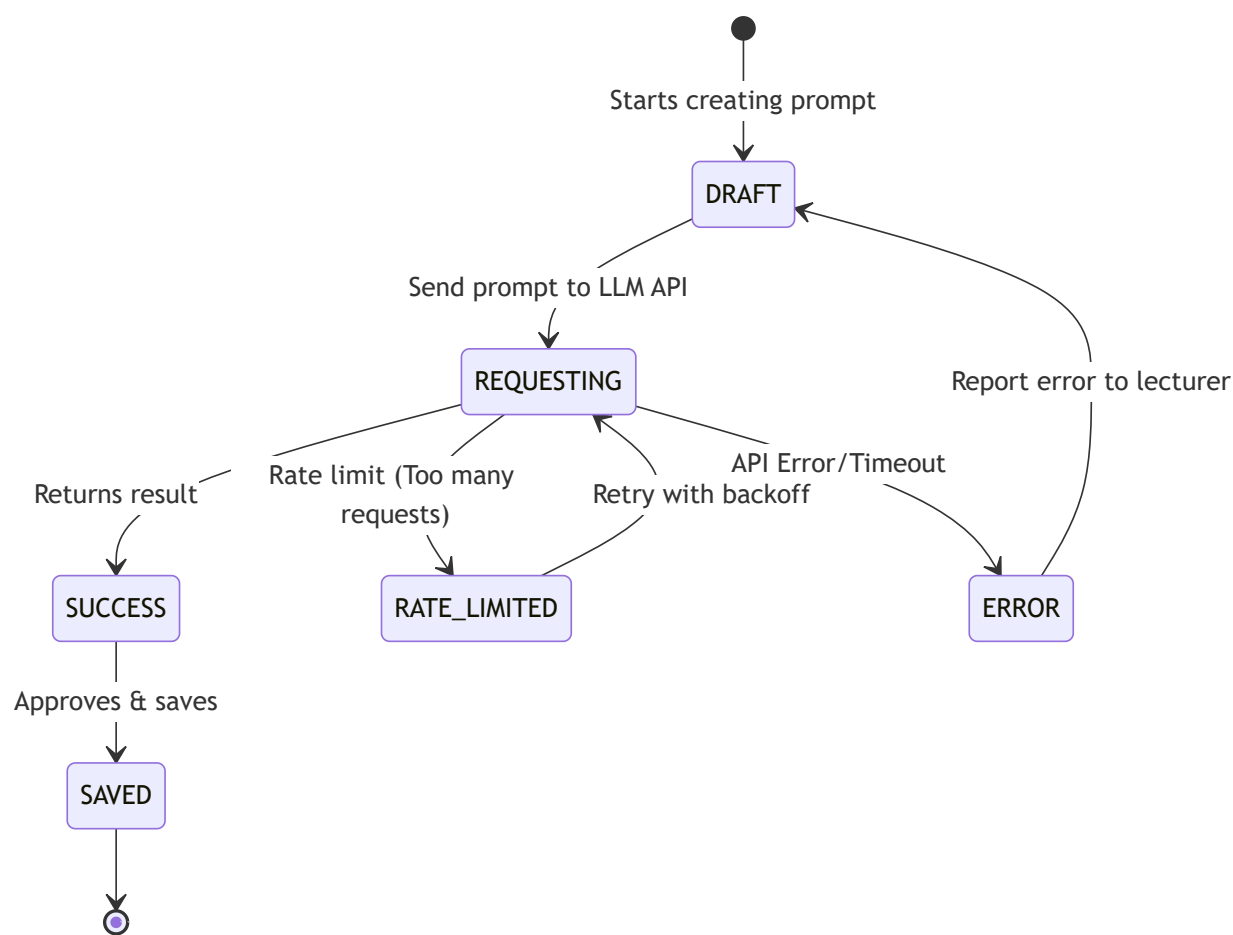


6.4. State Diagrams

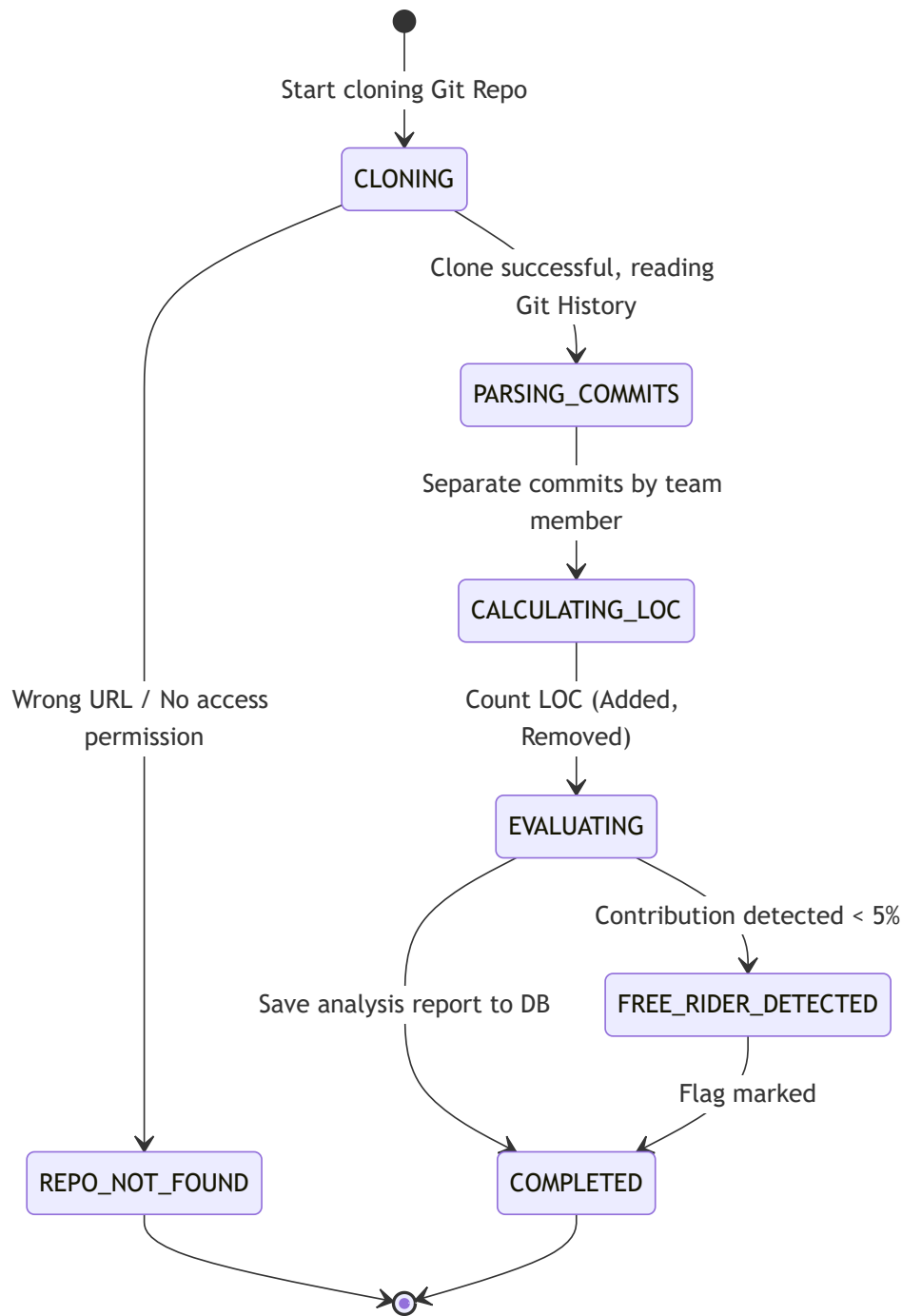
6.4.1. Submission Lifecycle



6.4.2. GenAI Request Lifecycle



6.4.3. Git Analytics Process Lifecycle



6.5. ERD (Entity-Relationship Diagram)

Design Note — Class Table Inheritance (CTI): The `USERS` table stores only attributes shared across all roles, while role-specific data is separated into child tables (`STUDENTS` , `LECTURERS` , `ADMINS`) linked via a 1-to-1 FK on `user_id` . This pattern ensures referential integrity at the database level — for example, `CLASSES.lecturer_id` can only reference a row that actually exists in `LECTURERS` , preventing accidental assignment of a student ID to a lecturer-only column. The `role` discriminator column is retained in `USERS` so the application layer knows which child table to JOIN without a multi-table probe.

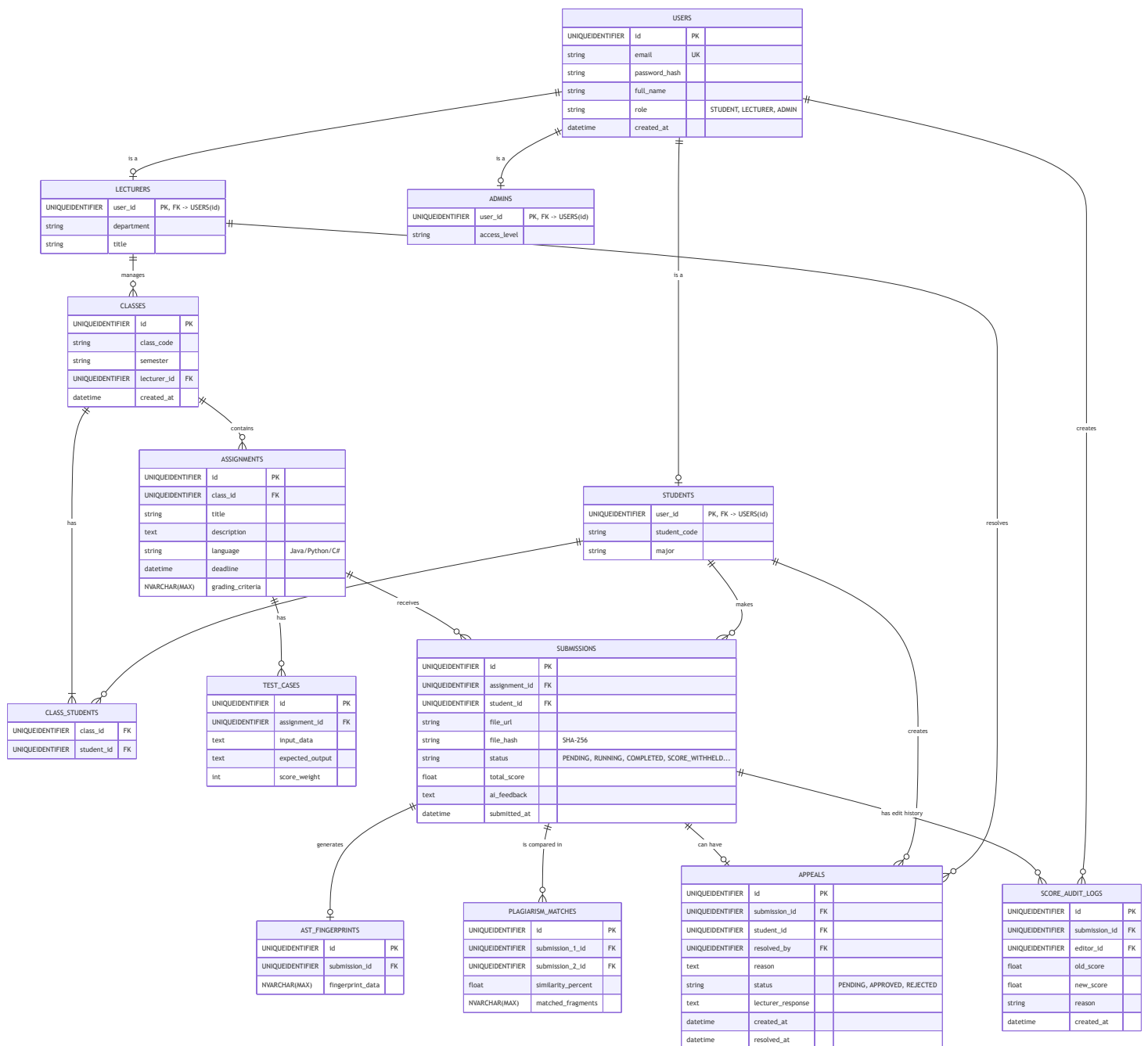


Table	Column	Data Type	Constraints	Description
USERS	id	UNIQUEIDENTIFIER	PK	Primary key, user identifier
	email	VARCHAR(255)	UNIQUE, NOT NULL	FPT email for login
	password_hash	VARCHAR(255)		Password (if using traditional login)

Table	Column	Data Type	Constraints	Description
	full_name	VARCHAR(100)	NOT NULL	Full name
	role	VARCHAR (ENUM)	NOT NULL	Role discriminator: STUDENT, LECTURER, ADMIN (determines which child table to JOIN)
	created_at	TIMESTAMP	DEFAULT NOW()	Account creation timestamp
STUDENTS	user_id	UNIQUEIDENTIFIER	PK, FK -> USERS(id)	1-to-1 with USERS
	student_code	VARCHAR(20)	UNIQUE	Student Roll Number (e.g., SE150000)
	major	VARCHAR(50)		Student's major (e.g., SE, AI, IA)
LECTURERS	user_id	UNIQUEIDENTIFIER	PK, FK -> USERS(id)	1-to-1 with USERS
	department	VARCHAR(100)		Department (e.g., Computing Fundamentals)
	title	VARCHAR(50)		Academic title (e.g., PhD, MSc)
ADMINS	user_id	UNIQUEIDENTIFIER	PK, FK -> USERS(id)	1-to-1 with USERS
	access_level	VARCHAR(50)		Admin role level (e.g., SuperAdmin, Moderator)
CLASSES	id	UNIQUEIDENTIFIER	PK	Primary key, class identifier

Table	Column	Data Type	Constraints	Description
	class_code	VARCHAR(50)	NOT NULL	Class code (e.g., SE1801)
	lecturer_id	UNIQUEIDENTIFIER	FK -> LECTURERS(user_id)	Lecturer in charge of the class
CLASS_STUDENTS	class_id	UNIQUEIDENTIFIER	FK -> CLASSES(id)	Foreign key to class
	student_id	UNIQUEIDENTIFIER	FK -> STUDENTS(user_id)	Foreign key to student
ASSIGNMENTS	id	UNIQUEIDENTIFIER	PK	Primary key, assignment identifier
	class_id	UNIQUEIDENTIFIER	FK -> CLASSES(id)	Foreign key to class
	title	VARCHAR(255)	NOT NULL	Assignment title
	description	TEXT		Assignment description
	language	VARCHAR(20)	NOT NULL	Programming language for Sandbox (Java, Python, C#)
	deadline	TIMESTAMP	NOT NULL	Submission deadline
	grading_criteria	NVARCHAR(MAX)		AI-generated grading criteria
TEST_CASES	id	UNIQUEIDENTIFIER	PK	Primary key, test case identifier
	assignment_id	UNIQUEIDENTIFIER	FK -> ASSIGNMENTS(id)	Belongs to which assignment
	input_data	TEXT		Input (StdIn)
	expected_output	TEXT		Expected output (StdOut)
	score_weight	INT	NOT NULL	Score weight

Table	Column	Data Type	Constraints	Description
SUBMISSIONS	id	UNIQUEIDENTIFIER	PK	Primary key, submission identifier
	assignment_id	UNIQUEIDENTIFIER	FK -> ASSIGNMENTS(id)	Belongs to which assignment
	student_id	UNIQUEIDENTIFIER	FK -> STUDENTS(user_id)	Submitter
	file_url	VARCHAR(255)	NOT NULL	File link .zip (S3/Local)
	file_hash	VARCHAR(64)		SHA-256 for integrity verification
	status	VARCHAR (ENUM)	NOT NULL	Status (PENDING, QUEUED...) (Mapped as string due to MSSQL limits)
	total_score	FLOAT		Total score
	ai_feedback	TEXT		AI Clean Code feedback
	submitted_at	TIMESTAMP	DEFAULT NOW()	Date and time of submission
AST_FINGERPRINTS	id	UNIQUEIDENTIFIER	PK	Primary key, AST fingerprint identifier
	submission_id	UNIQUEIDENTIFIER	FK -> SUBMISSIONS(id)	Fingerprint of which submission
	fingerprint_data	NVARCHAR(MAX)	NOT NULL	Winnowing k-gram hash array
PLAGIARISM_MATCHES	id	UNIQUEIDENTIFIER	PK	Primary key, match identifier
	submission_1_id	UNIQUEIDENTIFIER	FK -> SUBMISSIONS(id)	First submission

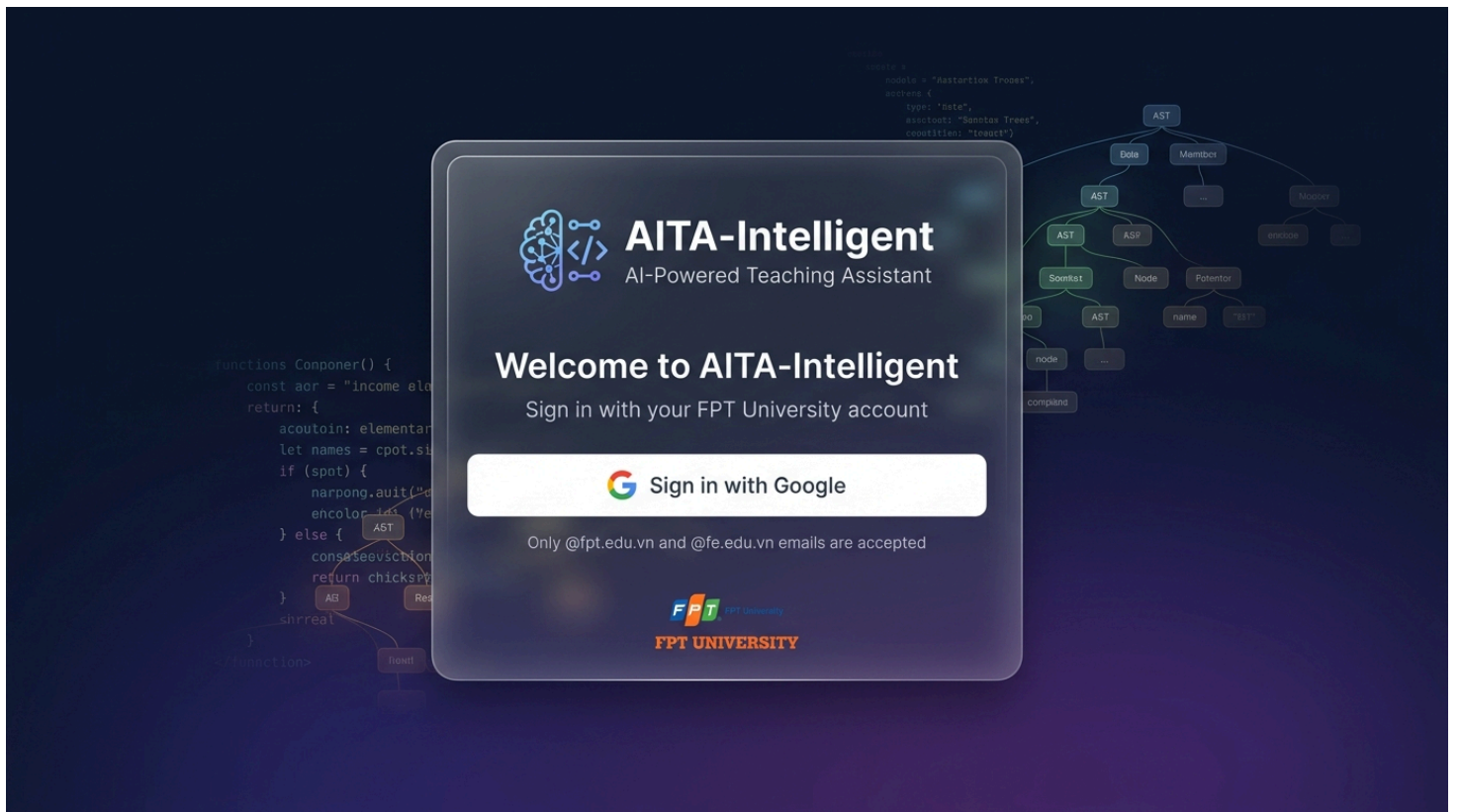
Table	Column	Data Type	Constraints	Description
	submission_2_id	UNIQUEIDENTIFIER	FK -> SUBMISSIONS(id)	Second submission
	similarity_percent	FLOAT	NOT NULL	Percentage of similarity (0-100)
	matched_fragments	NVARCHAR(MAX)		Array of matching line numbers/blocks
APPEALS	id	UNIQUEIDENTIFIER	PK	Primary key, appeal identifier
	submission_id	UNIQUEIDENTIFIER	FK -> SUBMISSIONS(id)	Appeal for which submission
	student_id	UNIQUEIDENTIFIER	FK -> STUDENTS(user_id)	Foreign key to student who created the appeal
	resolved_by	UNIQUEIDENTIFIER	FK -> LECTURERS(user_id), NULLABLE	Lecturer who resolved the appeal (NULL while PENDING)
	reason	TEXT	NOT NULL	Student's appeal reason
	status	ENUM	DEFAULT 'PENDING'	Status (PENDING, APPROVED, REJECTED)
	lecturer_response	TEXT		Lecturer's response
	created_at	TIMESTAMP	DEFAULT NOW()	Timestamp of creation
	resolved_at	TIMESTAMP		Timestamp of resolution
SCORE_AUDIT_LOGS	id	UNIQUEIDENTIFIER	PK	Primary key, log identifier

Table	Column	Data Type	Constraints	Description
	submission_id	UNIQUEIDENTIFIER	FK -> SUBMISSIONS(id)	Affected submission
	editor_id	UNIQUEIDENTIFIER	FK -> USERS(id)	Who made the edit (Lecturer or Admin — intentionally references USERS, not a child table)
	old_score	FLOAT		Previous score
	new_score	FLOAT		Updated score
	reason	TEXT	NOT NULL	Reason for score change
	created_at	TIMESTAMP	DEFAULT NOW()	Timestamp of edit

6.7. UI Wireframes & Screen Flow

This section describes the key screens of the AITA-Intelligent system, their layout, components, and the navigation flow between them.

6.7.1. Screen 1 — Login Page (Google SSO)

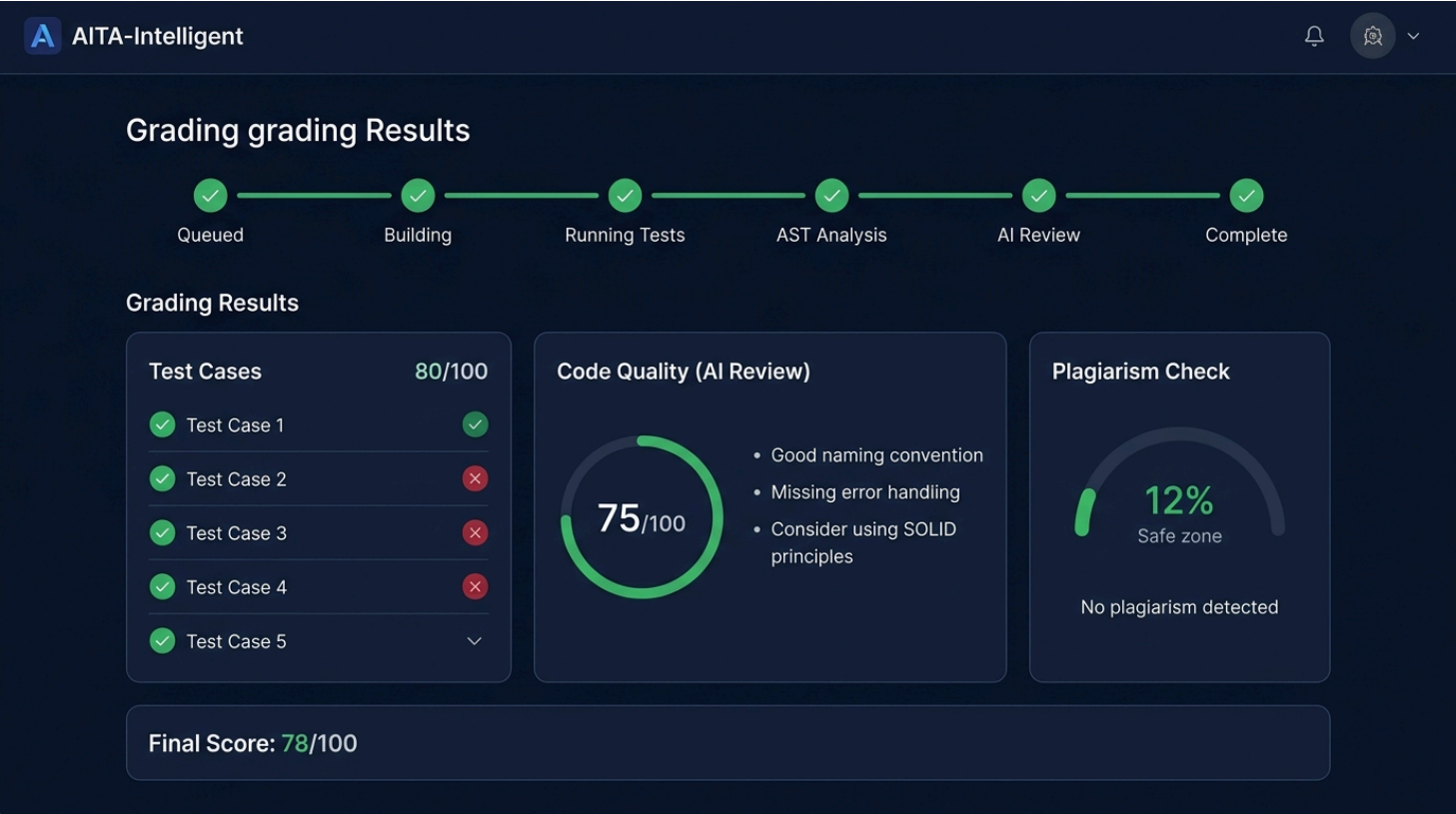


Attribute	Description
URL	/login
Layout	Centered card on gradient background
Components	FPT University logo, Application title "AITA-Intelligent", "Sign in with Google" button (primary CTA), Footer with version info
Actions	Click "Sign in with Google" → Google OAuth 2.0 consent screen → Redirect to /dashboard on success
Validation	Only @fpt.edu.vn and @fe.edu.vn email domains accepted. Others shown error toast.

6.7.2. Screen 2 — Student Dashboard

Attribute	Description
URL	/dashboard (role: Student)
Layout	Top navbar (avatar, notifications, logout) + Main content area with assignment cards
Components	- Assignment Cards: Title, Class code, Deadline countdown, Status badge (Not Submitted / Submitted / Graded) - Sort/Filter: By deadline (nearest first), by status - Quick Stats bar: Total assignments, Submitted count, Average score
Actions	Click card → Navigate to /assignments/:id/submit Click graded card → Navigate to /submissions/:id/results
Real-time	WebSocket listens for grading completion events → Status badge auto-updates without page reload

6.7.3. Screen 3 — Submission & Grading Results

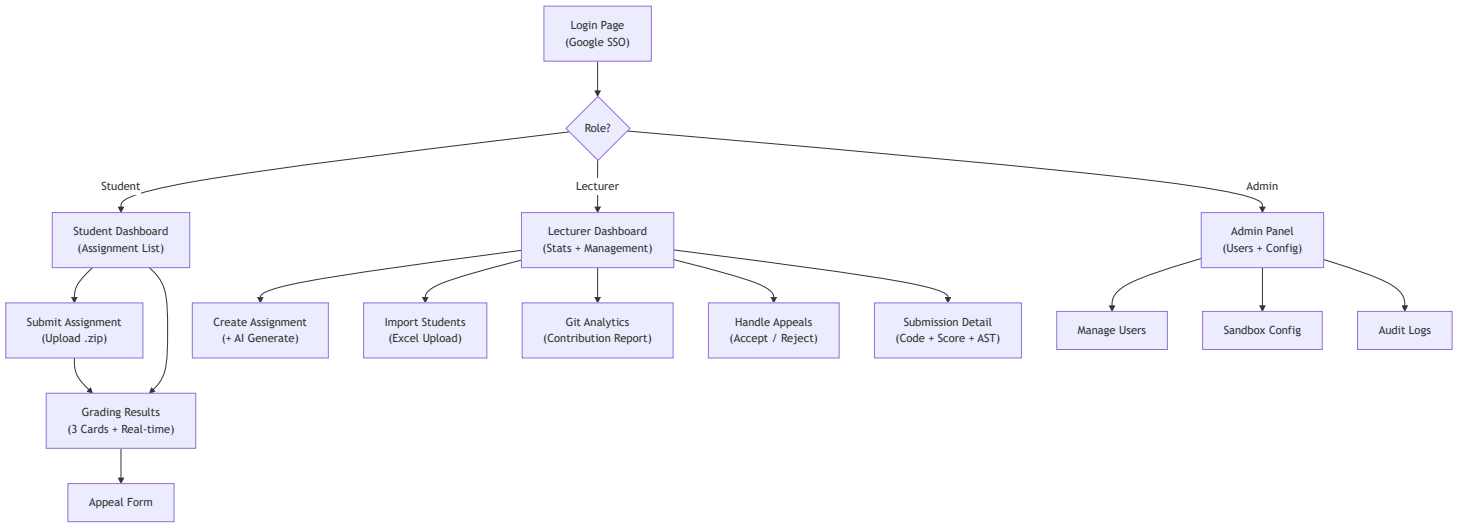
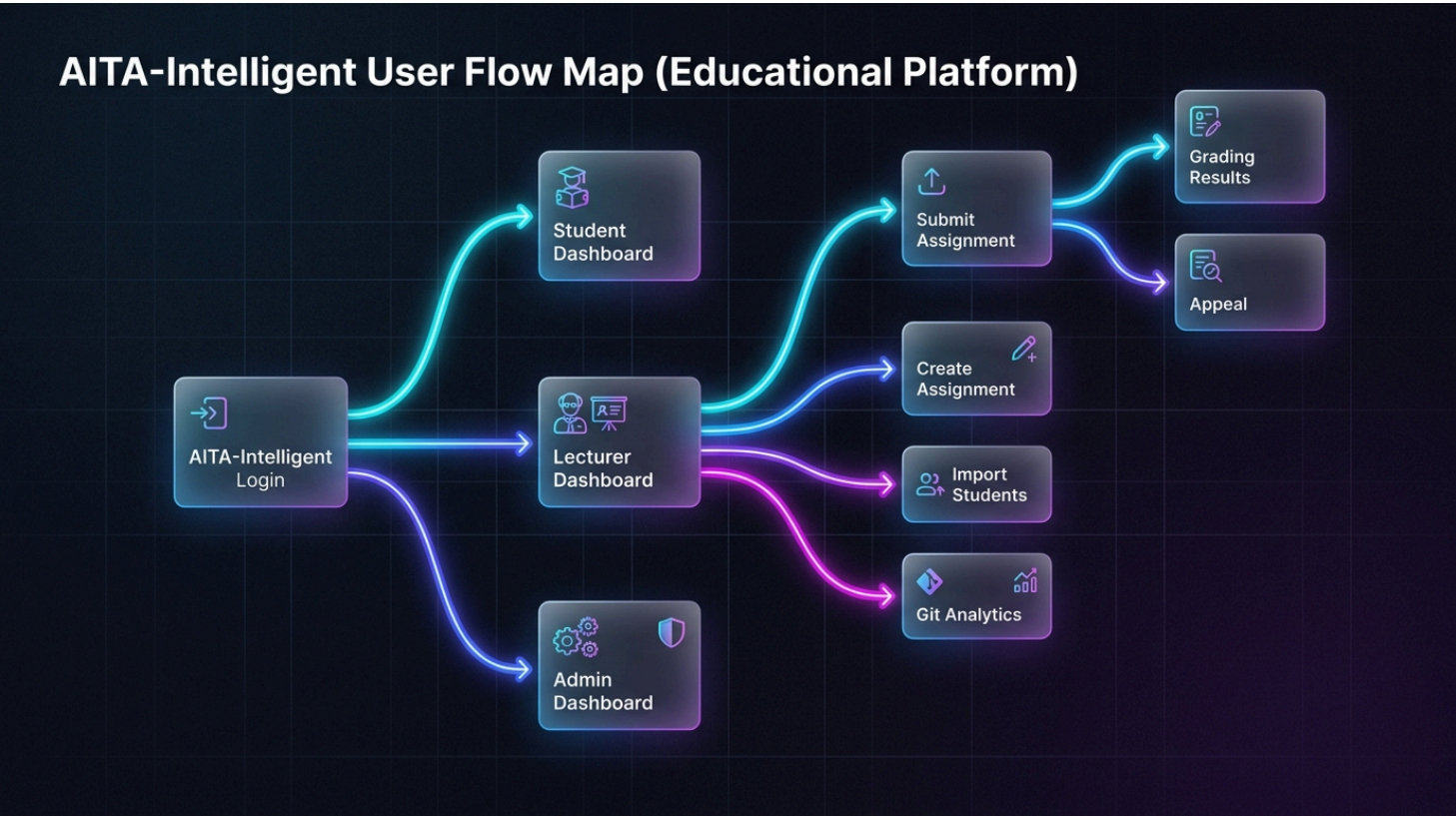


Attribute	Description
URL	/submissions/:id/results
Layout	Progress stepper (top) + 3 result cards (horizontal row) + Final score bar (bottom)
Components	
Progress Stepper	6 steps: Queued → Building → Running Tests → AST Analysis → AI Review → Complete. Active step: yellow spinner. Completed: green checkmark.
Card 1: Test Cases	Title + score (e.g., 80/100). List of test cases with ✓ (passed) / ✗ (failed). Expandable: Input, Expected Output, Actual Output.
Card 2: Code Quality (AI Review)	Circular gauge (0-100). AI feedback list: "✓ Good naming convention", "⚠ Missing error handling", "💡 Consider SOLID principles".
Card 3: Plagiarism Check	Semicircle gauge with color zones: 0-30% green (Safe), 30-60% yellow (Warning), 60-100% red (Danger). Match details if flagged.
Final Score Bar	"Final Score: XX/100" (large font). If status is SCORE_WITHHELD, display "Score pending lecturer review due to suspected plagiarism". "Appeal this result" button if student disagrees.
Real-time	WebSocket connection pushes step-by-step progress updates. All 3 cards render simultaneously on "Complete".

6.7.4. Screen 4 — Lecturer Dashboard



6.7.5. Screen Flow Diagram



7. Non-Functional Requirements

7.1. Security

ID	Requirement	Measurement Criteria
NFR-01	Code Execution Engine must completely isolate resources	RAM $\leq$ 512MB (--memory), limited CPU shares (--cpus), external network 100% disabled (--network=none) — self-configured via Docker Engine API

ID	Requirement	Measurement Criteria
NFR-02	Execution sandbox must defend against common malware	Block: fork bomb (<code>--pids-limit</code>), symlink escape (read-only filesystem <code>--read-only</code>), <code>/proc</code> mount (custom seccomp profile), disk exhaustion (<code>--storage-opt size= quota</code>) — all self-implemented and verified via dedicated security test cases (not standard unit tests)
NFR-03	JWT authentication must be stored in HttpOnly Cookie	Cookie inaccessible from JavaScript (XSS prevention)
NFR-04	Google SSO must only accept FPT email domains	Whitelist: <code>@fpt.edu.vn</code> , <code>@fe.edu.vn</code>
NFR-05	API keys (OpenAI, Gemini) must not be hard-coded	Store in environment variables (<code>.env</code>), never commit to Git
NFR-18	(P2P Defense) System must log every sandbox execution in detail	Log includes: <code>submission_id</code> , <code>start_time</code> , <code>end_time</code> , <code>exit_code</code> , <code>resource_usage</code> (RAM/CPU peak), <code>security_alert</code> . Retain for minimum 30 days.

7.2. Performance

ID	Requirement	Measurement Criteria
NFR-06	Average API response time	$\leq 200\text{ms}$ for standard APIs (CRUD)
NFR-07	End-to-end grading time per submission	≤ 90 seconds (includes: Code Execution + AST + AI). <i>Note: The 30s limit applies to code execution in Sandbox; 90s is the total time for the entire pipeline.</i>
NFR-08	System must handle concurrent load	≥ 100 requests/minute (stress test with k6 or JMeter)
NFR-09	Grading results must be cached	Redis cache for result APIs, TTL = 5 minutes, response $\leq 100\text{ms}$

7.3. Scalability

ID	Requirement	Measurement Criteria
NFR-10	Redis Queue must support batch grading	Process ≥ 50 concurrent submissions without blocking API gateway
NFR-11	System must support adding new programming languages	Build and register a new Docker image with the required language runtime + add ANTLR grammar for AST support

7.4. Reliability

ID	Requirement	Measurement Criteria
NFR-12	Bulk student import must use DB Transaction	Import all successfully OR rollback entirely (ACID)
NFR-13	Redis Queue must have retry mechanism	Max 3 retries with exponential backoff
NFR-14	Sandbox resources must be cleaned after completion	Code Execution Engine must explicitly destroy the container (<code>docker rm -f</code>) after execution or timeout; verified via Docker event listener, not third-party polling

7.5. Usability

ID	Requirement	Measurement Criteria
NFR-15	Responsive interface on both Web and Mobile	Support: Desktop (≥ 1024px), Tablet (≥ 768px), Mobile (≥ 375px)
NFR-16	Real-time result updates without page reload	Using WebSocket (Socket.IO) with API polling fallback
NFR-17	Compilation error explanations in Vietnamese	AI returns explanations in Vietnamese natural language

8. System Constraints

8.1. Technology Constraints

- **Frontend Web:** React 18+ with Vite, TailwindCSS
- **Backend:** Node.js 20+ with TypeScript, Prisma ORM
- **AST Engine:** Python 3.11+ with FastAPI, ANTLR4 (`antlr4-python3-runtime` for Java grammar)
- **Database:** Microsoft SQL Server 2019+
- **Message Queue:** Redis 7+ with BullMQ
- **Code Execution:** Custom Docker-based sandbox orchestrated via Docker Engine API (e.g., Dockerode for Node.js)
- **CI/CD:** GitHub Actions
- **Cloud Deploy:** Azure / AWS / Vercel (team's choice)

8.2. MVP Scope Limitations

The following features are scoped as **simplified/secondary** for the 10-week MVP. Full implementations are planned for future iterations.

Feature	MVP Scope	Future Scope
AST Plagiarism Detection	Java only (ANTLR parser)	Add Python (<code>ast</code> module), C# (Roslyn)
GenAI Assignment Generation (US-08)	Basic draft generation (assignment description + sample test cases)	Full rubric auto-generation, multi-round refinement
Git Analytics (US-11)	Basic metrics: commits count, LOC per member	PR analysis, contribution timeline charts, automated free-rider scoring
Code Execution Engine	Custom Docker containers with basic resource limits (memory, CPU, network, PID)	Stronger isolation via gVisor or Firecracker microVMs; per-language pre-warmed container pools for lower latency
Mobile App	Responsive web (PWA-ready)	Native React Native / Expo app

8.3. Risk Assessment

#	Risk	Impact	Mitigation
R-01	Resubmission Policy — No clear rule on how many times a student can resubmit before deadline	Students may flood the queue with unlimited resubmissions	Limit to max 5 submissions per assignment per student. Show remaining attempts on UI.
R-02	API Rate Limiting — OpenAI/Gemini API may throttle or reject requests during peak hours	AI Code Review step fails, resulting in incomplete grading	Implement queue-based retry with exponential backoff. Fallback: skip AI review and mark as "AI Review Pending".
R-03	Audit Trail for Manual Edits — Lecturer manually overrides a score but no log is kept	Disputes during P2P Defense cannot be resolved with evidence	Log every manual score edit: <code>editor_id</code> , <code>old_score</code> , <code>new_score</code> , <code>reason</code> , <code>timestamp</code> . Immutable audit table.
R-04	False-Positive Plagiarism — AST + Winnowing flags legitimate code as plagiarism (e.g., boilerplate, starter code)	Students unfairly penalized, appeals increase	Allow lecturer to whitelist specific code patterns/files. Display similarity breakdown (which functions matched). Student can appeal with explanation.
R-05	Data Retention Policy — No defined policy for how long submission files, logs, and personal data are stored	GDPR/PDPA compliance risk, storage costs grow unbounded	Define retention: Submission files = 1 semester, Sandbox logs = 30 days (NFR-18), Personal data = until account deletion. Auto-purge scripts run monthly.
R-06	Custom Sandbox Security Gaps — Self-built Docker isolation may miss edge cases (e.g., kernel-level exploits, container escape) that a mature third-party sandbox would already handle	Security vulnerability could allow malicious code to affect the host server	Follow Docker security hardening checklist (non-root user, read-only FS, drop all capabilities except required, seccomp/AppArmor profile); schedule a dedicated security review sprint before demo; document known limitations in the Risk Assessment for P2P Defense transparency

8.4. Business Constraints

- The system only serves students and lecturers belonging to FPT University (verified by email domain).
- Each submission is limited to a maximum of 10MB (.zip file).
- Maximum code execution time is 30 seconds per submission (enforced by the Code Execution Engine's own timeout controller, which force-kills the container via `docker kill`).
- OpenAI/Gemini API key uses a single key with exponential backoff retry.

8.5. Project Constraints

- Development timeline: 10 weeks.
- Development team: 4-6 students.
- Code must pass linting (ESLint/Prettier for TypeScript, Flake8 for Python).
- Unit test coverage $\geq 80\%$ on core modules (scoring, validation, matching logic - excluding live Docker sandbox/AI integration paths).

End of SRS Document – Version 1.2 (Updated per Milestone 1 Feedback + Contradiction Fix)

Development Team: SWP391 – Group 3